

Class 11: AlphaFold

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Here we read the results from AlphaFold and try to interpret all the models and quality score metrics:

```
library(bio3d)

pth <- "dimer_23119/"
pdb.files <- list.files(path=pth, full.names = TRUE, pattern=".pdb")
```

Align and superpose all these models

```
file.exists(pdb.files)
```

```
[1] TRUE TRUE TRUE TRUE TRUE
```

```
pdb.s <- pdbaln(pdb.files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
dimer_23119//dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000.pdb
dimer_23119//dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_5_seed_000.pdb
dimer_23119//dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_4_seed_000.pdb
dimer_23119//dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_1_seed_000.pdb
dimer_23119//dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....
```

Extracting sequences

```

pdb/seq: 1   name: dimer_23119//dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_1
pdb/seq: 2   name: dimer_23119//dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_2
pdb/seq: 3   name: dimer_23119//dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_3
pdb/seq: 4   name: dimer_23119//dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_4
pdb/seq: 5   name: dimer_23119//dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_5

```

```

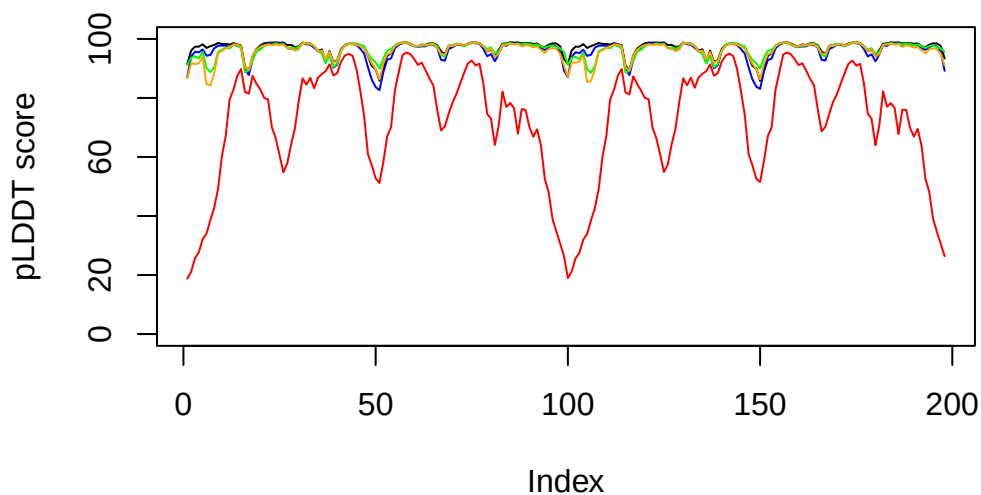
library(bio3dview)
#view.pdbs(pdbs)

```

```

plot(pdbs$b[1,], typ="l", ylim=c(0,100), ylab="pLDDT score")
lines(pdbs$b[2,], typ="l", col="blue")
lines(pdbs$b[3,], typ="l", col="green")
lines(pdbs$b[4,], typ="l", col="orange")
lines(pdbs$b[5,], typ="l", col="red")

```



Predicted Alignment Error for domains

```

library(jsonlite)

```

```
# Listing of all PAE JSON files
pae_files <- list.files(path=pth,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

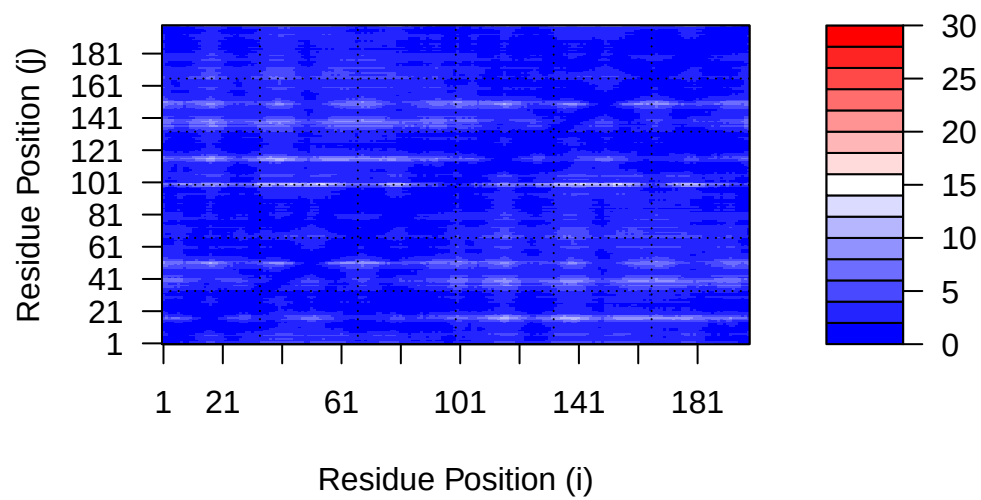
attributes(pae1)
```

```
$names
[1] "plddt"    "max_pae" "pae"      "ptm"      "iptm"
```

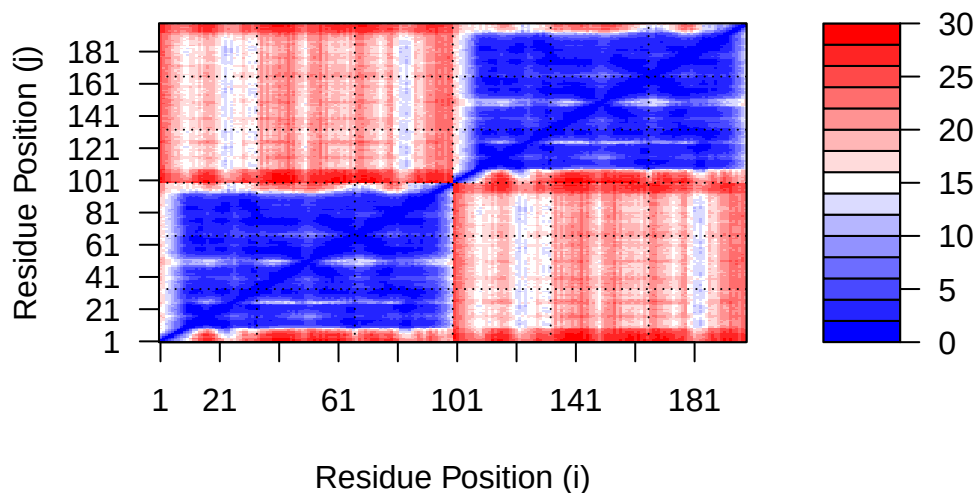
```
# Per-residue pLDDT scores
# same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.44 96.06 97.38 97.38 98.19 96.94
```

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



```
plot.dmat(pae5$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



```
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 198 non NA positions

```
rd
```

```

dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000
dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_5_seed_000
dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_4_seed_000
dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_1_seed_000
dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000
dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000
dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_5_seed_000
dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_4_seed_000
dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_1_seed_000
dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000
dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000
dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_5_seed_000
dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_4_seed_000

```

```
dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_1_seed_000
dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000
```

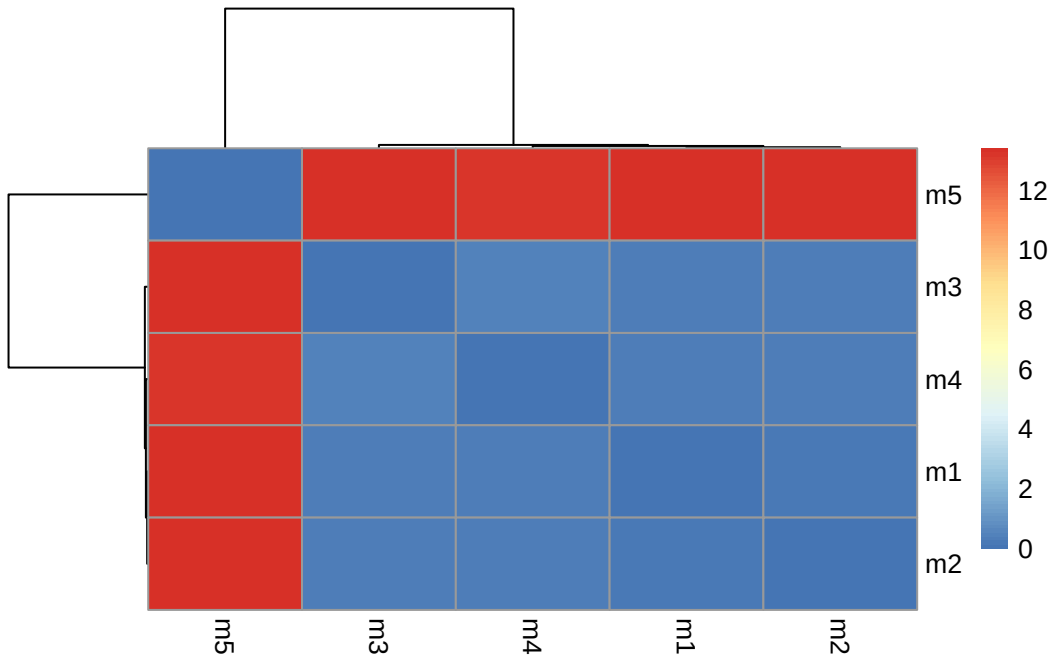
dimer_23119_unrelaxed.

```
dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000
dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_5_seed_000
dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_4_seed_000
dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_1_seed_000
dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000
```

dimer_23119_unrelaxed.

```
dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000
dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_5_seed_000
dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_4_seed_000
dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_1_seed_000
dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000
```

```
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



Score Residue conservation from alignment file

```
aln_file <- list.files(path=pth,  
                      pattern=".a3m$",  
                      full.names = TRUE)  
aln_file
```

```
[1] "dimer_23119//dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

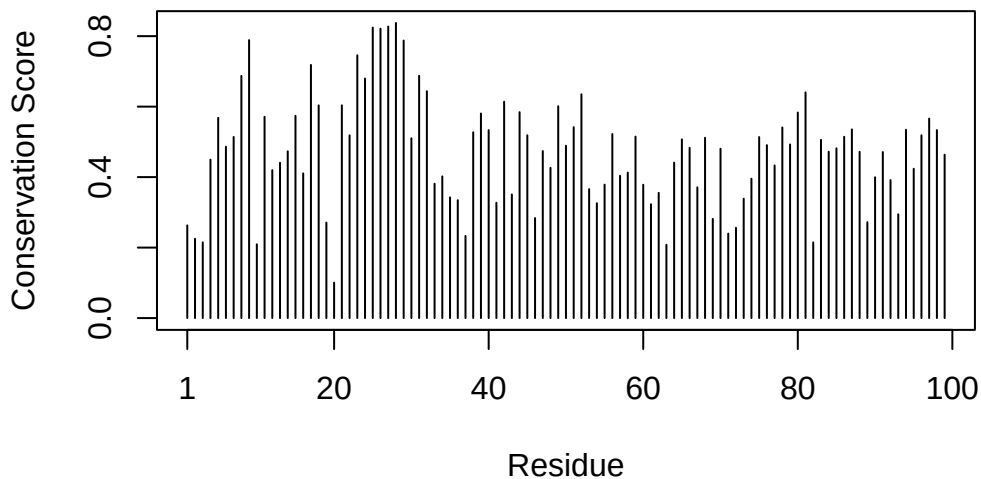
```
[1] " ** Duplicated sequence id's: 101 **"  
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5378 132
```

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99],  
       ylab="Conservation Score")
```



Find the consensus sequence at a very high cutoff

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb.files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```